Git and GitHub

By

Tanvir Hussain

INTRODUCTION

Git and GitHub are two of the most commonly used terms in the world of software development. Have you ever wondered what exactly they are and how they work? In this blog, let's explore Git and GitHub, understand their differences, and learn about their uses.

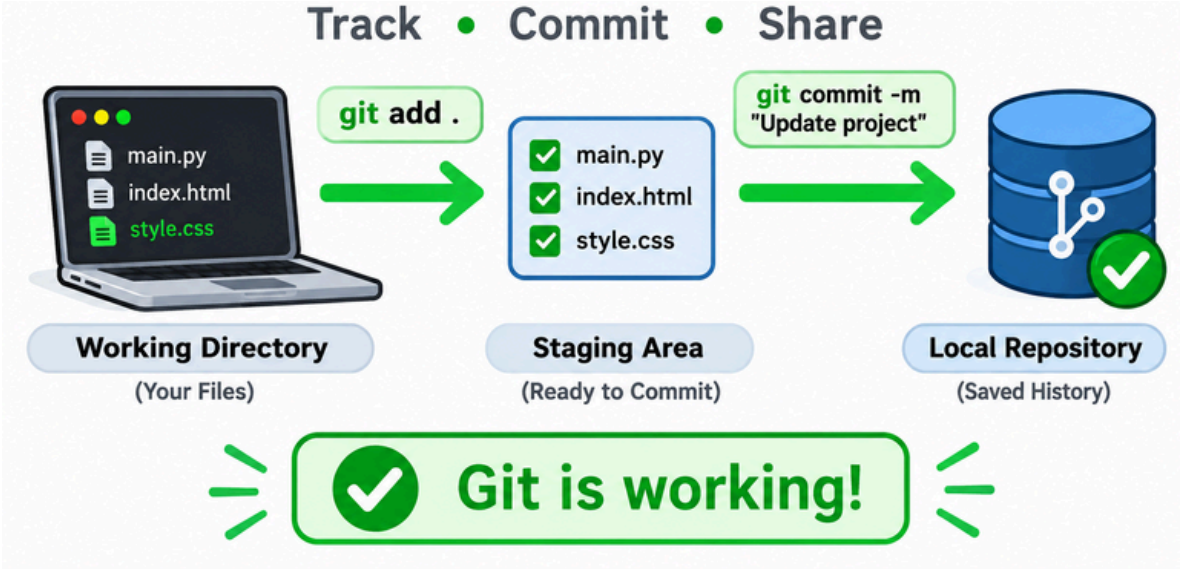
GIT

The Linux operating system is extremely popular among developers. It was created by Linus Torvalds. During the development of the Linux kernel, Linus faced challenges in managing contributions from multiple developers. Keeping track of changes, identifying who made them, and maintaining a stable codebase became increasingly difficult.

To solve these problems, Linus Torvalds developed Git in 2005. It quickly became one of the most widely used version control systems in the world. This brought him a huge success.

Git is a free and open-source Version Control System (VCS) developed by Linus Torvalds in 2005. It allows developers to track changes made to their source code over time. It helps developers save previous versions of their code and return to older versions whenever required. Git also allows developers to work offline and make commits (save changes in the local repository). These changes can later be pushed to a remote server whenever an internet connection is available. This helps developers manage their projects and collaborate efficiently without worrying about losing their previous work.

We can download and use Git on various operating systems, including Windows, macOS, and Linux distributions.



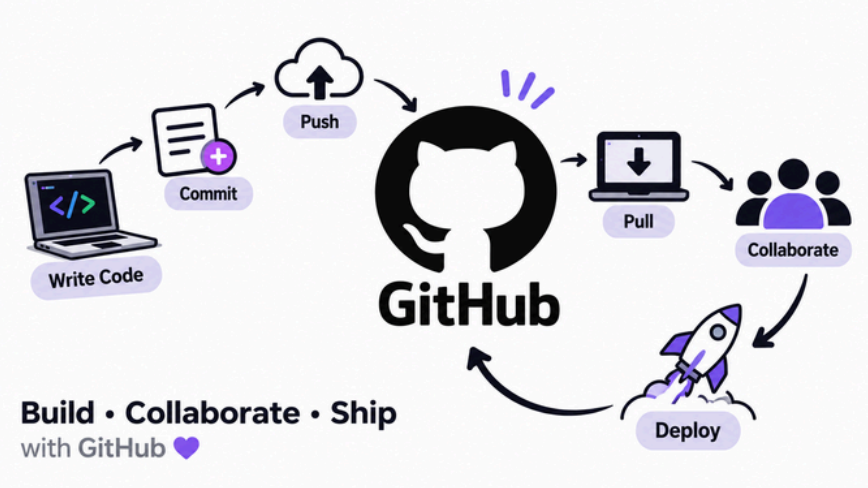
GITHUB

While Git is a version control system, GitHub is a hosting platform where we can host our Git repositories. A repository can also be called a project folder, but technically, it contains more than just files. It also stores the complete version history tracked by Git.

We can fetch and push our code to GitHub very easily. GitHub provides several features that make software development and collaboration easier.

We can use GitHub for the following purposes:

- Host repositories.
- Push changes to the source code.
- Fetch or pull changes from remote repositories.
- Host static websites for free using GitHub Pages.
- Collaborate with other developers.
- Create and track issues.



Are Git and GitHub the same?

Now, you may think that Git and GitHub are the same. But that's not true!

Although they are closely related, they have different purposes.

Git is just like a personal assistant who takes care of your codebase or repository. It tracks all the changes, including who made them and when they were made. It also allows developers to return to previous versions of their code whenever necessary.

GitHub, on the other hand, is a hosting platform where we can store our repositories online. Think of it as a drawer where you can keep your files along with their complete records and share them with others.

In simple words, Git manages and tracks your code, while GitHub helps you host and collaborate on it.



A repository is like a drawer where you keep your files looked after by someone